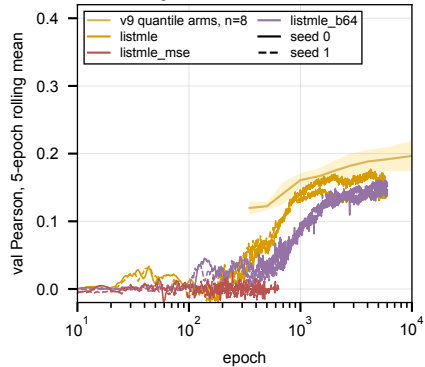
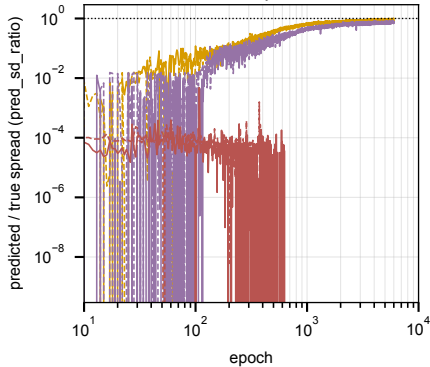


a

ranking arms vs the incumbent band

**b**

scale, which ListMLE only bounds from below

**c**

the objective on validation

